Loss functions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Keep each loss attached to its cases and definition.

Show signed residuals,squares and all-case divisor;distinguish supplied baselines,training evidence and untouched final evaluation.

Keep one habit: Assess both independent metrics and the changed ranking with checked keys;avoid invented fitting provenance.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary **target**: Calculate per-example **squared error** and **mean squared error**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

No. We also need a **target** or other evaluation criterion and appropriate examples.

Opening retrieval

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish architectural choices from parameter values

Reference example, not the only acceptable wording: Teacher explanation: There are 8 weights and 2 biases, total 10.

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

Look for: Trace how one layer supplies the next layer **input**s

Reference example, not the only acceptable wording: Teacher explanation: Scores are -2 and 2; activations are 0 and 2; y = 0 - 2 + 1 = -1.

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate a dot product by matching feature positions

Reference example, not the only acceptable wording: Teacher explanation: Contributions are 6 and -5, so the score is 1.

Pictured end goal

By the end, I can explain loss functions using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Parameters and architecture

Composing layers

Dot products and weighted sums

Keep paired **prediction** and **target** rows visible in both 45-minute sessions. Require **residual**s, per-row penalties and the denominator before accepting an average.

This week uses regression and ordinary MSE; do not introduce a different scaling convention midway.

Outcomes

Calculate per-example **squared error** and **mean squared error**

Distinguish **residual** cancellation from a low loss

Separate **training** loss from independent evaluation

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach pairing,penalty and denominator as one calculation

> Keep the main,pictured and independent numerical fixtures separate.

Main definitions,complete comparison and guided key

A loss assigns a numerical penalty to a **prediction** compared with a **target**. For this regression lesson, **residual** e = **prediction** - **target**, and **squared error** is e squared.

Squaring makes both positive and negative **residual**s contribute nonnegative penalties. We use ordinary **mean squared error** without a one-half factor.

The **mean squared error**, or MSE, is (1 + 1 + 4) / 3 = 2. It summarises these three examples, not every possible future **input**. If **target**s use cm, **residual**s use cm and MSE uses square cm.

For **target**s [1, 3] and **prediction**s [2, 2], the mean **residual** is zero because 1 and -1 cancel. MSE is 1, correctly recording that both **prediction**s miss.

A mean **residual** can describe directional tendency, but it is not a substitute for an error magnitude.

The guard prevents silently ignoring an unmatched example or dividing by zero. It still assumes paired, numeric, valid records. Wrong row alignment produces a meaningless comparison even when list lengths agree.

Same target vector[1, 3, 2] · Predictions · Residuals prediction–target · Squared errors · MSE over 3

> **Same target vector[1, 3, 2]:** P
>
> **Predictions:** [2, 2, 4]
>
> **Residuals prediction–target:** [1, –1, 2]
>
> **Squared errors:** [1, 1, 4]
>
> **MSE over 3:** 2

> **Same target vector[1, 3, 2]:** Q
>
> **Predictions:** [1, 2, 2]
>
> **Residuals prediction–target:** [0, –1, 0]
>
> **Squared errors:** [0, 1, 0]
>
> **MSE over 3:** 1/3

> **Same target vector[1, 3, 2]:** Fixed comparison baseline

> **Predictions:** [2, 2, 2]
>
> **Residuals prediction–target:** [1, –1, 0]
>
> **Squared errors:** [1, 1, 0]
>
> **MSE over 3:** 2/3

> **Same target vector[1, 3, 2]:** Supported R
>
> **Predictions:** [0, 3, 3]
>
> **Residuals prediction–target:** [–1, 0, 1]
>
> **Squared errors:** [1, 0, 1]
>
> **MSE over 3:** 2/3

The fixed[2, 2, 2]comparison vector is supplied for loss arithmetic; its training-fit history is not given. Do not infer that 2 was fitted from these evaluation targets. Pair predictions and targets by case identity, not only by list position after an independent sort. The opposite residual sign used in some earlier lessons gives the same squared/absolute losses, but here signed residual means prediction–target.

Keep the declared sign beside each calculation.

Guided: **residual**s [-1, 0, 1], squares [1, 0, 1], MSE 2/3, tied with the baseline under MSE.

Exit: zero **training** MSE establishes an exact fit on those recorded **target**s under the chosen loss; it does not establish performance on unseen cases or correct labels.

Complete separate picture facts and scope

Compare candidate losses with the same **target**s and denominator.

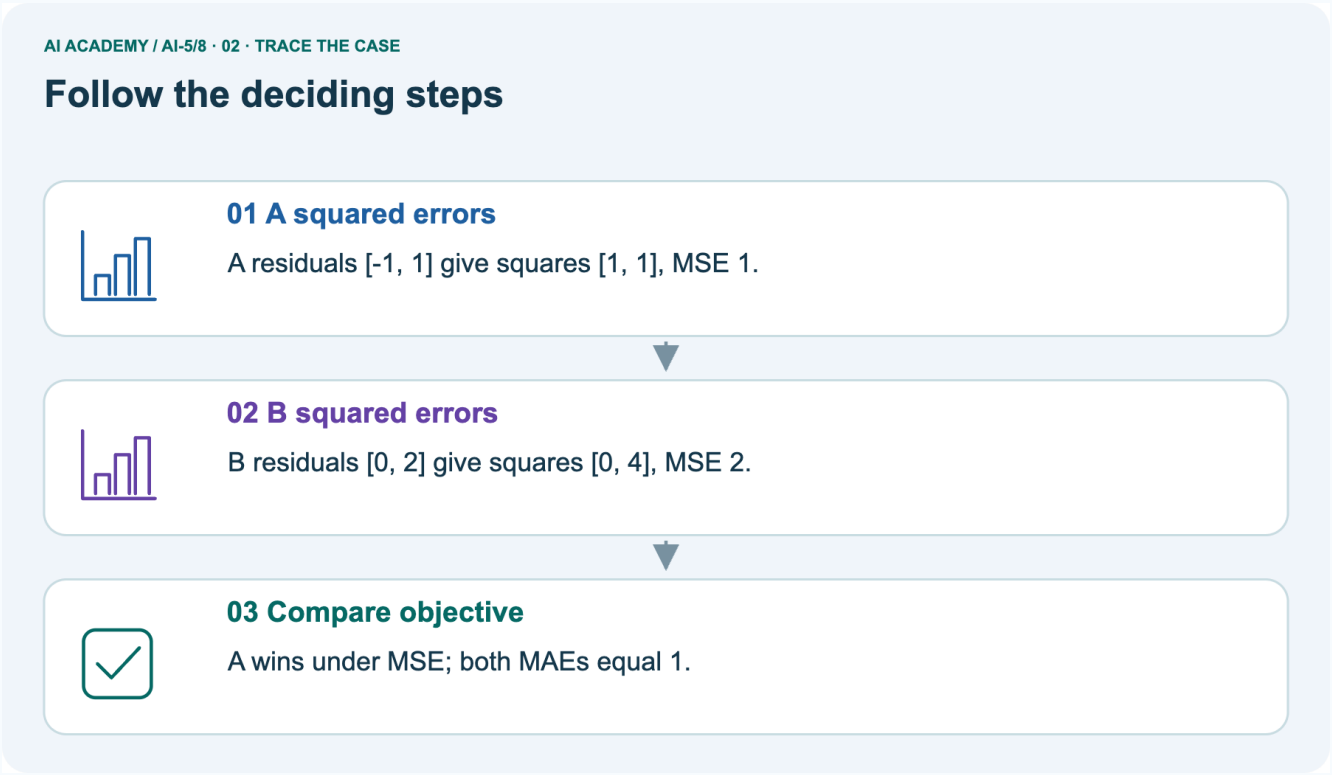
Targets [1, 3]; A predicts [0, 4]; B predicts [1, 5]. Use MSE over both cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate two-case fixture:targets[1, 3], A[0, 4]has residuals[–1, 1], MSE 1; B[1, 5]has[0, 2], MSE 2. Both MAEs are 1. Squaring gives the one error of size 2a penalty 4, while the two errors of size 1 give total 2. The choice A under MSE becomes a tie under MAE. Do not substitute the main three-case P/Q/Rvectors or the selected independent task's inputs.

These supplied prediction pairs are arithmetic evidence, not claimed training runs.

Trace MSE-versus-MAEranking



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. A squared errors	A residuals [-1, 1] give squares [1, 1], MSE 1.
2. B squared errors	B residuals [0, 2] give squares [0, 4], MSE 2.
3. Compare objective	A wins under MSE; both MAEs equal 1.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

For this loss calculation, the signed error is **prediction** minus **target**. MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case.

Squaring makes the larger individual error count more. State the objective before selecting a model, because these two objectives give different comparisons.

Training loss is not automatically a held-out performance estimate or a complete measure of use consequences.

Change: Choose MAE instead of MSE as the sole criterion.

Prediction: A and B tie at MAE 1.

Reason: **Squared error** penalises B’s concentrated error more, while absolute errors both sum 2.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Target	The recorded value a model is intended to predict
Residual	Prediction minus target under the sign convention used here
Squared error	The square of a numerical residual
Mean squared error	The average squared error over the evaluated examples

Use the same targets and denominator

CANDIDATE PREDICTIONS	SQUARED ERRORS	MSE
P is [2, 2, 4]	[1, 1, 4]	6 divided by 3 is 2
Q is [1, 2, 2]	[0, 1, 0]	1 divided by 3
Baseline is [2, 2, 2]	[1, 1, 0]	2 divided by 3

Each row evaluates a separate candidate against targets [1, 3, 2]. Q has the lowest MSE on these three cases. The comparison alone does not establish performance on unseen examples.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Use the same targets and denominator

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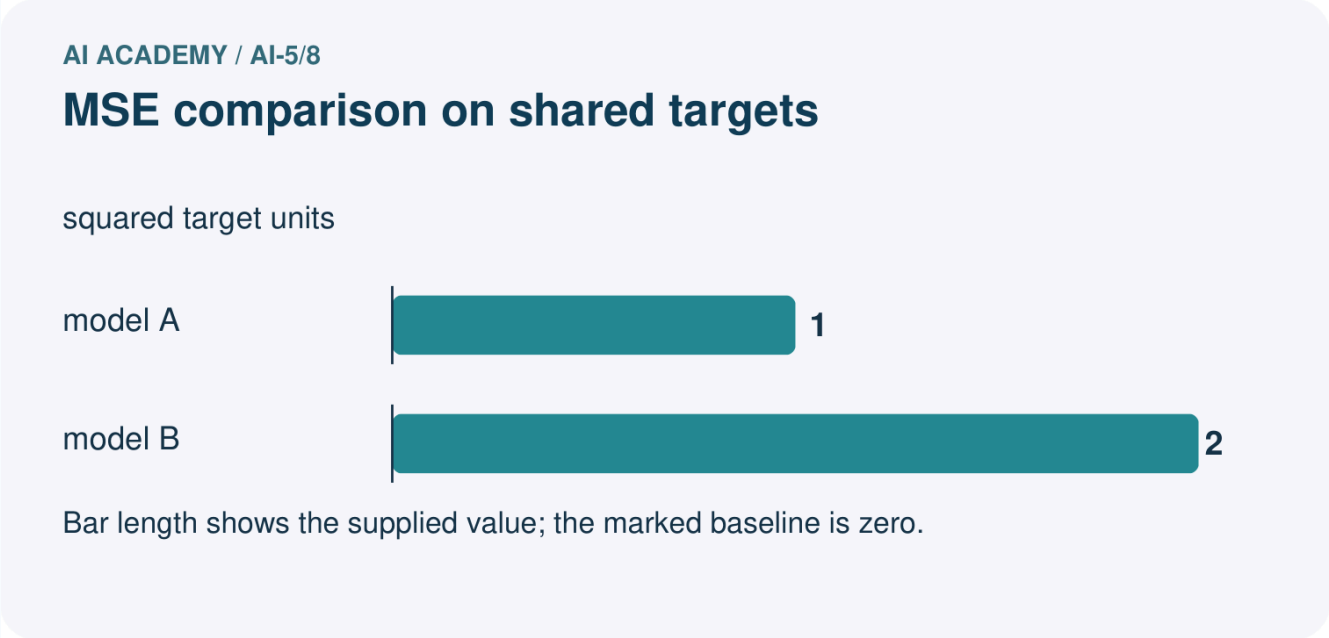
> **Candidate R predicts [0, 3, 3] for targets [1, 3, 2]. Calculate residuals, squared errors and MSE. Compare R with the baseline.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Loss functions

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/8

Loss functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. A squared errors

A residuals [-1, 1] give squares [1, 1], MSE 1.

↓

2. B squared errors

B residuals [0, 2] give squares [0, 4], MSE 2.

↓

3. Compare objective

A wins under MSE; both MAEs equal 1.

CHANGE ONE THING

Choose MAE instead of MSE as the sole criterion.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: **Target**s [0, 4] have **prediction**s [2, 2]. Compare mean **residual** and MSE.

Teacher check: Teacher explanation: **Residual**s 2 and -2 average to 0; squares 4 and 4 give MSE 4. Both **prediction**s are wrong despite zero mean **residual**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Loss functions. Proposed explanation: “Average **residual** is the loss”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Signed **residual**s can cancel. Apply the specified penalty before averaging.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short **target**ed group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

The same **prediction**s produce a strict A win under MSE and a tie under MAE; “better” depends on the declared objective. Changed-case check: A and B tie at MAE 1.

Squared error penalises B’s concentrated error more, while absolute errors both sum 2.

Changed-case reasoning: A and B tie at MAE 1. **Squared error** penalises B’s concentrated error more, while absolute errors both sum 2.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a **residual** and squared-error table and permit a calculator for fractions.

Extension: if both **target**s and **prediction**s are converted from cm to mm, explain why **residual**s multiply by 10 and MSE by 100 while candidate ranking stays the same.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve why outputs alone do not establish correctness.

Time or stage	Teacher action
7 to 18 minutes	Calculate A, B and C row by row. Explain the sign convention and units.
18 to 32 minutes	Complete W1 and W2, showing how cancellation hides error.
32 to 45 minutes	Run or trace the guarded MSE code; try unequal and empty lists.
Session 2 start to 8 minutes	Use W3 to recover the full denominator, then compare the visual candidates.
8 to 20 minutes	Complete the guided R case and discuss squared versus absolute penalties.
20 to 35 minutes	Collect W4 independently, including both metrics and a qualified conclusion.
35 to 45 minutes	Use W5 and the exit response. Administer the fresh retry after feedback.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main P2, Q1/3, baseline/R2/3; W2 cancellationMSE4 not0. W3 denominator3 gives10/3. Pairing exampleMSE1 versuswrongly paired 5. Independent targets[0, 2, 4]:A[1, 2, 3]MSE 2/3, B[0, 2, 2]MSE 4/3; A wins MSE, bothMAE2/3 tie. Numerical references and future coverage still limit claims. Report baseline provenance as absent for the supplied constant comparison, not fitted on evaluation targets.

Diagnose the current misconceptions

Average **residual** is the loss: Signed **residual**s can cancel. Apply the specified penalty before averaging.

Ignore perfect **prediction**s in the average: The denominator includes every valid evaluated pair, including zero-error cases.

One metric gives the universal winner: Different losses express different penalties. Choose and disclose the criterion.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: **Target**s are [0, 2, 4]. Model A predicts [1, 2, 3]; model B predicts [0, 2, 2]. Calculate both MSEs and identify the better model under MSE on these cases. Calculate mean absolute errors too.

State what changes in the conclusion.

Teacher explanation: A **residual**s [1, 0, -1] give MSE 2/3; B [0, 0, -2] gives MSE 4/3, so A is better under MSE. Both mean absolute errors are 2/3, giving a tie under that metric. Neither ranking establishes future performance.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Calculate a loss

For **target**s [2, 4] and **prediction**s [1, 6], record **residual**s, squares and MSE.

Teacher explanation: **Residual**s are -1 and 2, squares 1 and 4, MSE 2.5.

W2 Expose cancellation

Targets [0, 4] have **prediction**s [2, 2]. Compare mean **residual** and MSE.

Teacher explanation: **Residual**s 2 and -2 average to 0; squares 4 and 4 give MSE 4. Both **prediction**s are wrong despite zero mean **residual**.

W3 Check the denominator

A learner totals **squared error**s 1, 0, 9 and divides by 2 because there are two nonzero errors. Correct the MSE.

Teacher explanation: All three evaluated examples count. MSE is 10 / 3, not 5. A zero-error example is still an example.

W5 Protect final testing

A team tries ten models and picks the one with the lowest final-test MSE. Can that same result be presented as an untouched final evaluation?

Teacher explanation: No. The test informed selection and became development **evidence**. Preserve the result with that disclosure, freeze the chosen **process** and use fresh independent evaluation for a new final claim.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: **residual**s [-1, 0, 1], squares [1, 0, 1], MSE 2/3, tied with the baseline under MSE.

Exit: zero **training** MSE establishes an exact fit on those recorded **target**s under the chosen loss; it does not establish performance on unseen cases or correct labels.

W1 Calculate a loss

W2 Expose cancellation

W3 Check the denominator

W4 Compare an unfamiliar evaluation independently

Targets are [0, 2, 4]. Model A predicts [1, 2, 3]; model B predicts [0, 2, 2]. Calculate both MSEs and identify the better model under MSE on these cases. Calculate mean absolute errors too.

State what changes in the conclusion.

Teacher explanation: A **residual**s [1, 0, -1] give MSE 2/3; B [0, 0, -2] gives MSE 4/3, so A is better under MSE. Both mean absolute errors are 2/3, giving a tie under that metric. Neither ranking establishes future performance.

W5 Protect final testing

Independent assessment and fresh retry

W4 requires MSEs 2/3 and 4/3, tied MAEs 2/3, and limited claims. Fresh retry: **target**s [1, 5], **prediction**s [3, 4] give **residual**s [2, -1], MSE 2.5 and MAE 1.5. Reject averaging the **residual**s to 0.5 as an error magnitude.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled loss functions **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate per-example **squared error** and **mean squared error**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide a **residual** and squared-error table and permit a calculator for fractions.

Extension: if both **target**s and **prediction**s are converted from cm to mm, explain why **residual**s multiply by 10 and MSE by 100 while candidate ranking stays the same.

Week 9 measures local changes in a simple loss curve.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/9, ai-5/11, ai-5/14. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching **target**: Calculate per-example **squared error** and **mean squared error**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: **Target**s [0, 4] have **prediction**s [2, 2]. Compare mean **residual** and MSE.

Reasoned practice answer: Teacher explanation: **Residual**s 2 and -2 average to 0; squares 4 and 4 give MSE 4. Both **prediction**s are wrong despite zero mean **residual**.

Check every learner before discussion: Use the worked case for Loss functions. Proposed explanation: “Average **residual** is the loss”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Signed **residual**s can cancel. Apply the specified penalty before averaging.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Choose MAE instead of MSE as the sole criterion. Require a **prediction** and a reason before comparing. Expected result: A and B tie at MAE 1.

Reason: **Squared error** penalises B’s concentrated error more, while absolute errors both sum 2.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

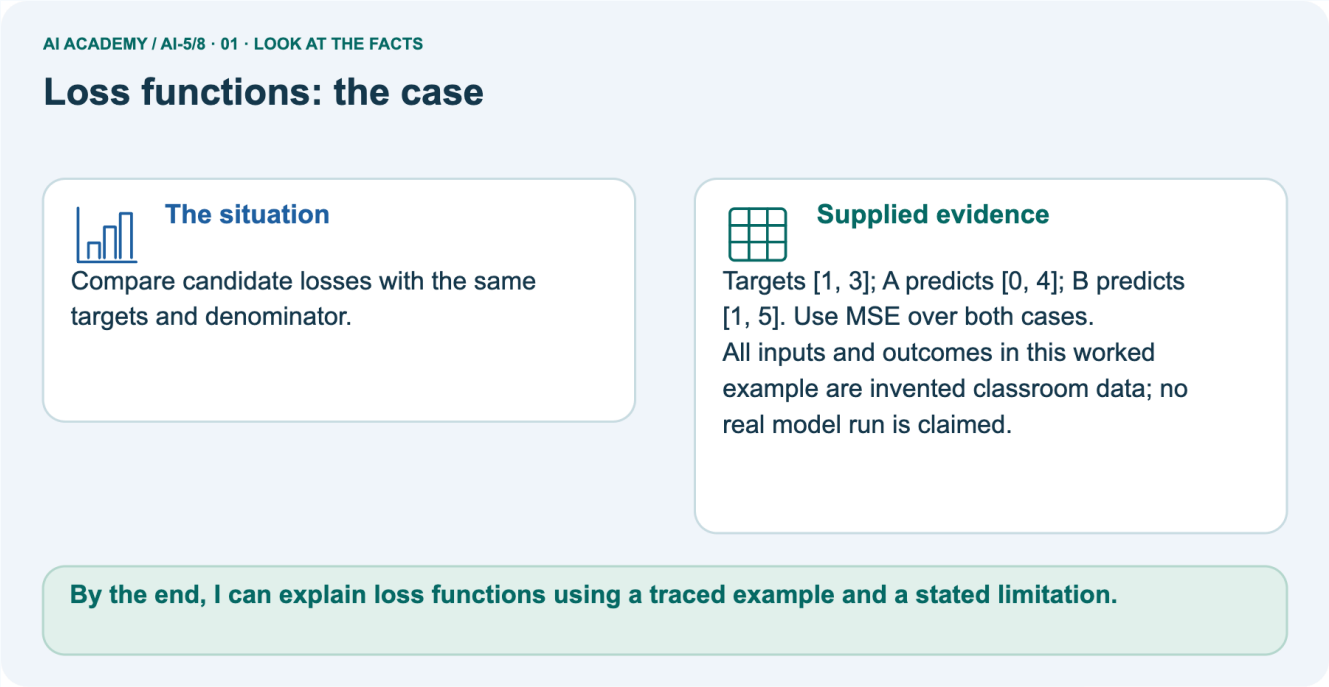
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Loss functions: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

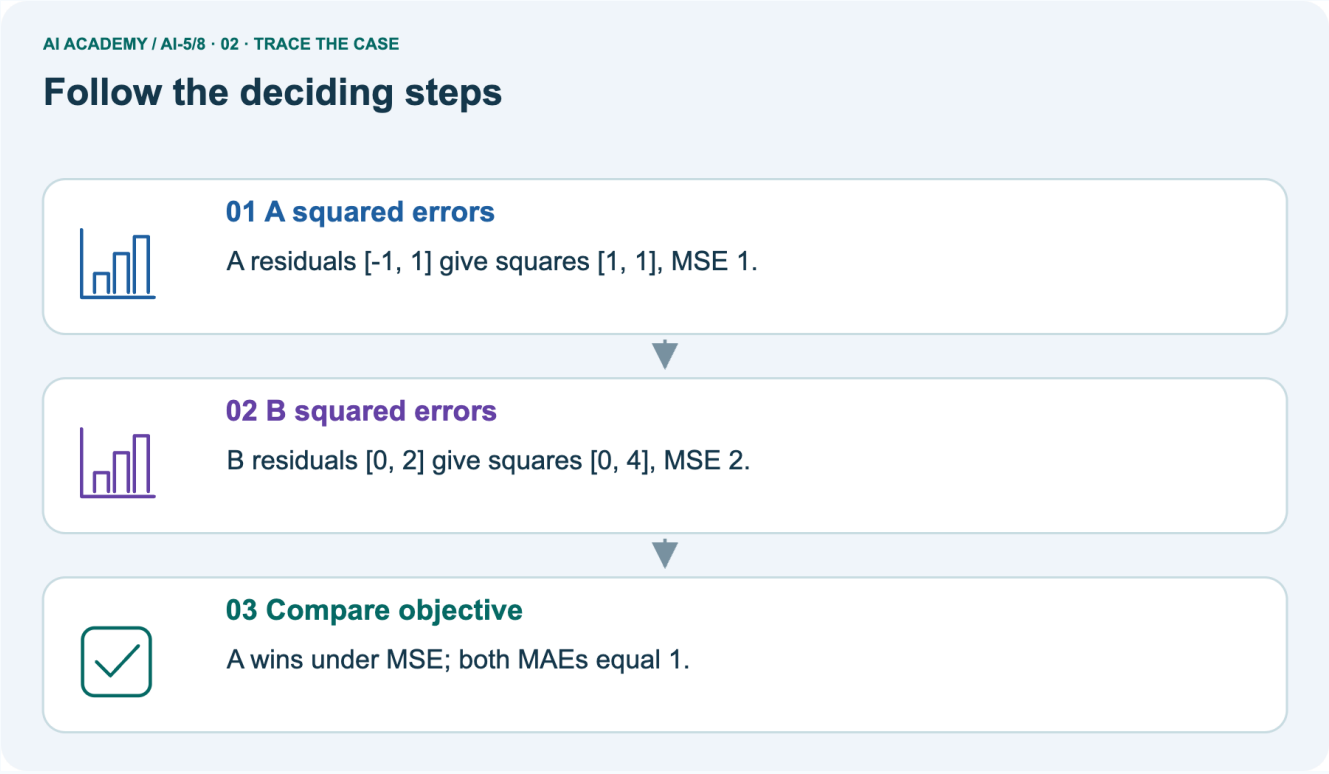
Compare candidate losses with the same **target**s and denominator. Supplied facts: **Target**s [1, 3]; A predicts [0, 4]; B predicts [1, 5]. Use MSE over both cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



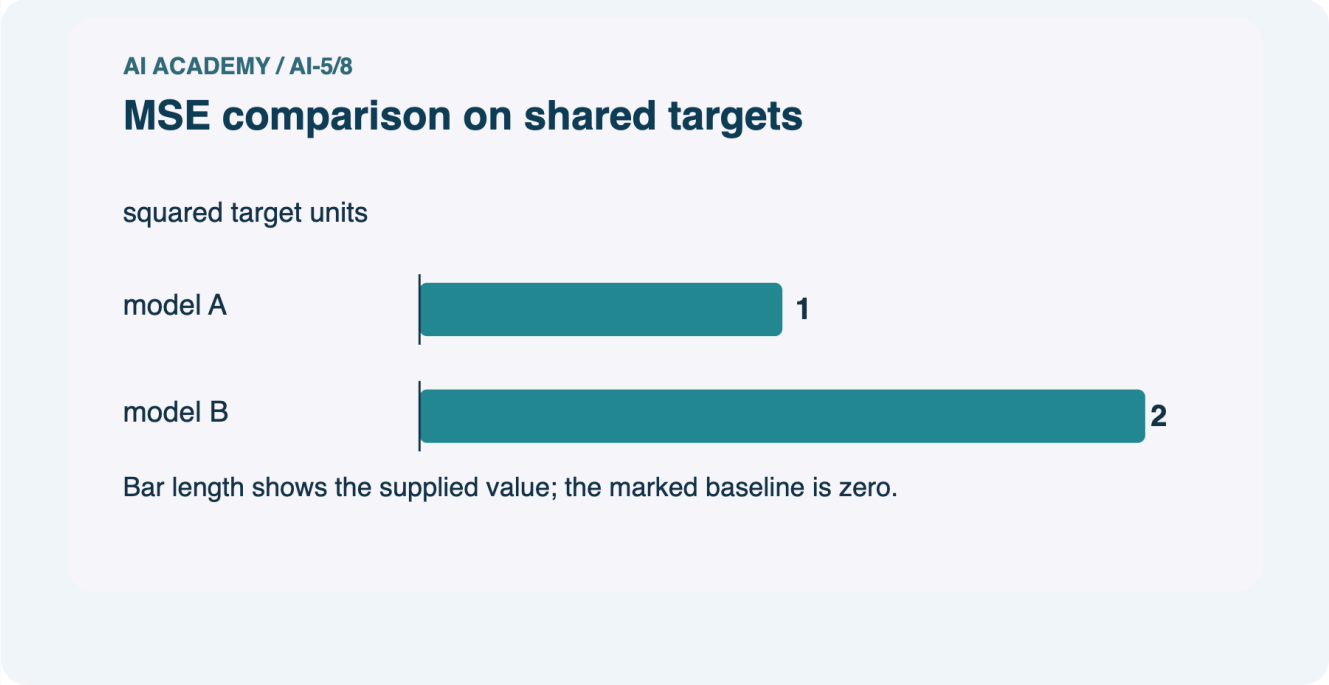
Follow the labelled picture with the matching explanation. Open the vector diagram

A **squared error**s: A **residual**s [-1, 1] give squares [1, 1], MSE 1. → B **squared error**s: B **residual**s [0, 2] give squares [0, 4], MSE 2. → Compare objective: A wins under MSE; both MAEs equal 1.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. MSE comparison on shared targets



Follow the labelled picture with the matching explanation. Open the vector diagram

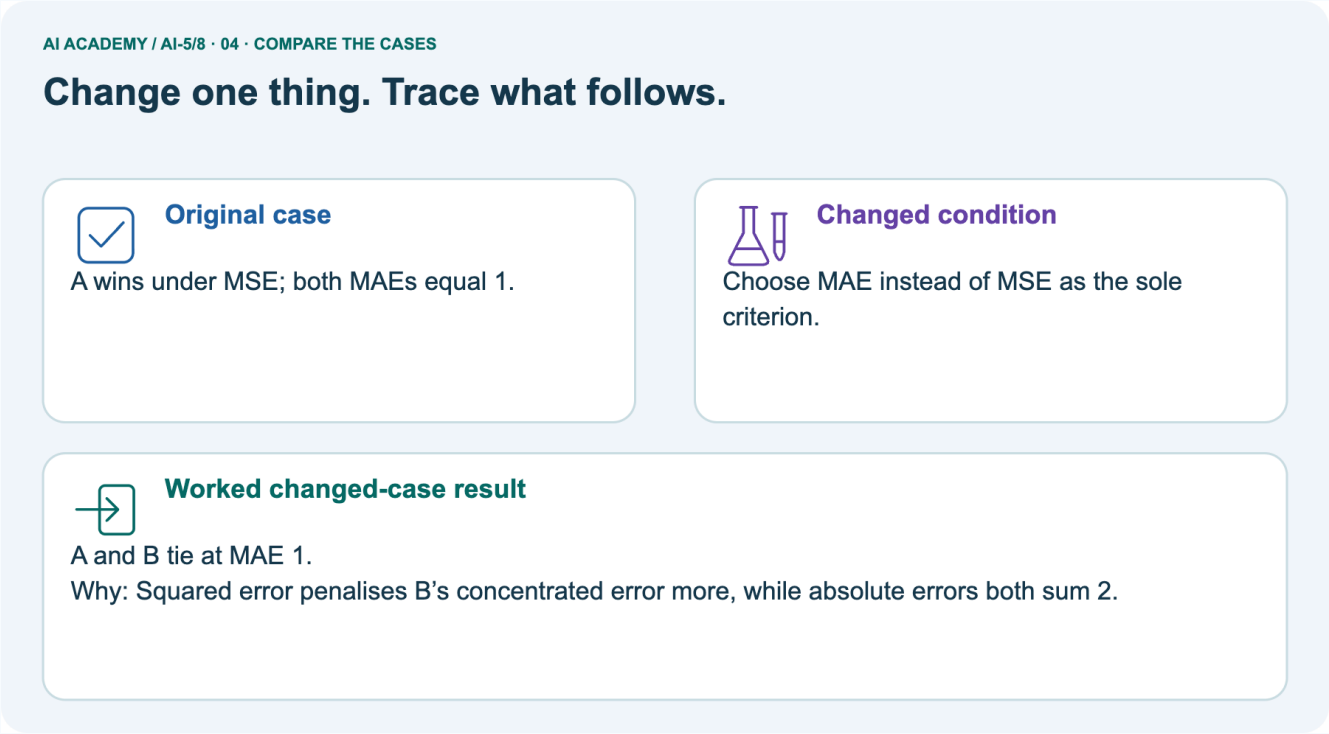
MSE comparison on shared **target**s. Units: squared **target** units. model A: 1. model B: 2.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A wins under MSE; both MAEs equal 1. Changed condition: Choose MAE instead of MSE as the sole criterion. Expected result: A and B tie at MAE 1.

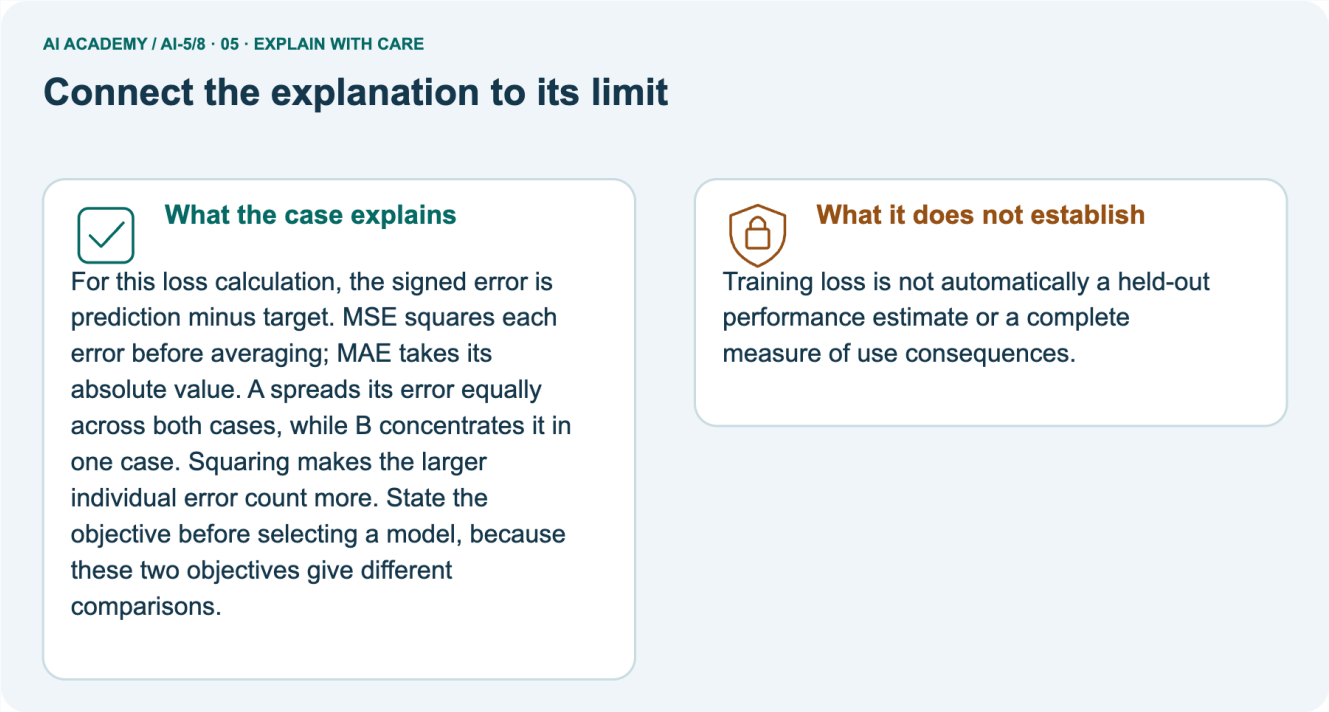
Reason: **Squared error** penalises B's concentrated error more, while absolute errors both sum 2.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: **Squared error** penalises B's concentrated error more, while absolute errors both sum 2.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

For this loss calculation, the signed error is **prediction** minus **target**. MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case.

Squaring makes the larger individual error count more. State the objective before selecting a model, because these two objectives give different comparisons.

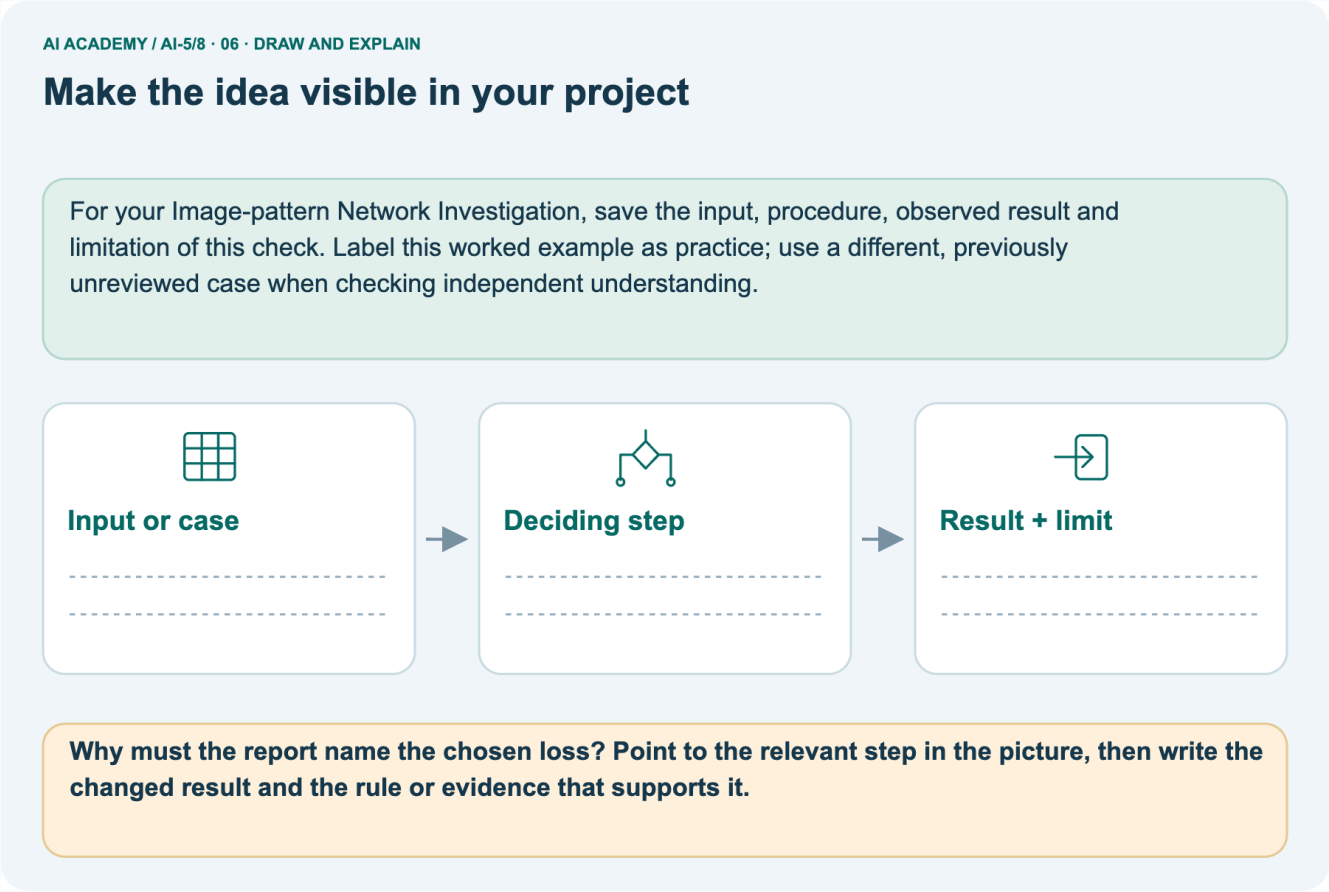
Limit: **Training** loss is not automatically a held-out performance estimate or a complete measure of use consequences.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Training** loss is not automatically a held-out performance estimate or a complete measure of use consequences.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The same **prediction**s produce a strict A win under MSE and a tie under MAE; “better” depends on the declared objective. Changed-case check: A and B tie at MAE 1.

Squared error penalises B’s concentrated error more, while absolute errors both sum 2.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Loss functions: the case

02. Follow the deciding steps

03. MSE comparison on shared targets

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.